

A Log-Chowla Quantifier Firewall

Frozen Squarefree Truncations, a Slow Diagonal, and Why They Do Not Prove the Actual Fixed-Two H3 Gate

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Abstract

The determinant-two pullback of TPC-127 exposes the literal sign $\lambda(n-2)\lambda(n)$, inviting comparison with the proved logarithmically averaged two-point Chowla theorem. We make that comparison with exact quantifiers. Tao's theorem fixes the two affine forms and, for any $\omega(x) \rightarrow \infty$, $1 \leq \omega(x) \leq x$, controls the terminal reciprocal window

$$\sum_{x/\omega(x) < m \leq x} \frac{\lambda(a_1 m + b_1)\lambda(a_2 m + b_2)}{m} = o(\log \omega(x)).$$

Using the finite CRT expansion of TPC-128, we obtain an unconditional corollary for every *fixed* determinant-two pair and every fixed squarefree cutoff R . A standard diagonal lemma lets a countable frozen family grow at some non-effective arbitrarily slow rate. We then prove exact Tauberian and finite-window counterexamples showing why terminal reciprocal cancellation does not supply an ordinary-weight all-prefix bound. It also gives no uniform rate in the actual growing moduli, origins, cutoffs, masks, phases, physical weights, and outer keys. The fixed-form theorem is genuine but auxiliary; it is not an L2 result for the physical family. The paper ends with the precise stronger statement that would be callable by H3.

1 The proved theorem with its native quantifiers

We record only the specialization needed here.

Theorem 1.1 (Logarithmically averaged two-point Chowla, specialized). *Fix $a_1, a_2 \in \mathbb{N}$ and $b_1, b_2 \in \mathbb{Z}_{\geq 0}$ with*

$$a_1 b_2 - a_2 b_1 \neq 0.$$

Let $\omega(x) \rightarrow \infty$ satisfy $1 \leq \omega(x) \leq x$. Then

$$\sum_{x/\omega(x) < m \leq x} \frac{\lambda(a_1 m + b_1)\lambda(a_2 m + b_2)}{m} = o(\log \omega(x)). \quad (1)$$

The forms and their coefficients are fixed before $x \rightarrow \infty$.

Proof. This is the Liouville specialization of the two-point logarithmically averaged Chowla–Elliott theorem [1]. $\square$

Remark 1.2 (No silent enlargement). [Theorem 1.1](#) is a terminal logarithmic-window statement. It is not stated here as a full interval $[1, x]$, an ordinary Cesàro average, a power-rate estimate, an all-prefix maximum, or a theorem uniform in growing coefficients. Those changes require separate arguments.

2 A rigorous frozen CRT corollary

Fix once and for all positive affine forms

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2,$$

with the primitive relations

$$(a, s) = 1, \quad (as, 2) = 1, \quad (d, s) = (u, a) = 1.$$

For fixed R , put

$$Q_R(m) = \sum_{\substack{k \leq R \\ k^2 | m}} \mu(k).$$

Theorem 2.1 (Fixed-form, finite-mask logarithmic cancellation). *For every fixed choice of a, s, d, u, R above and every admissible $\omega(x)$,*

$$\sum_{x/\omega(x) < z \leq x} \frac{Q_R(D(z))Q_R(V(z))\lambda(D(z))\lambda(V(z))}{z} = o(\log \omega(x)). \quad (2)$$

Changing or growing a, s, d, u, R with x is not included in this statement.

Proof. TPC-128 gives the finite identity

$$Q_R(D(z))Q_R(V(z)) = \sum_{\substack{k, \ell \leq R \\ (k, s) = (\ell, a) = (k, \ell) = 1}} \mu(k)\mu(\ell) \mathbf{1}_{z \equiv r_{k, \ell} \pmod{k^2 \ell^2}}$$

[2]. Fix one nonempty residue and write

$$z = r + Mt, \quad M = k^2 \ell^2.$$

The two forms become

$$D(r + Mt) = D(r) + sMt, \quad V(r + Mt) = V(r) + aMt,$$

whose coefficient determinant is

$$(sM)V(r) - (aM)D(r) = M\{sV(r) - aD(r)\} = 2M \neq 0.$$

After shifting the nonnegative t -origin by a fixed amount if necessary, [theorem 1.1](#) applies to these fixed forms.

Moreover,

$$\frac{1}{r + Mt} = \frac{1}{M} \frac{1}{t} + O_M(t^{-2})$$

away from finitely many initial terms. The total contribution of the error over any terminal window is $O_M(1)$, hence $o(\log \omega(x))$. Fixed affine changes of the two endpoints alter only $O_M(1)$ boundary terms and replace ω by an asymptotically comparable admissible terminal ratio. Summing the finitely many k, ℓ proves (2). $\square$

Remark 2.2 (What is unconditional here). [Theorem 2.1](#) is a valid analytic consequence of a proved theorem. It controls a frozen, reciprocal-weight, finitely truncated auxiliary family. It neither estimates the TPC physical multiplier nor lets the squarefree cutoff tend to infinity at a specified rate.

3 The only legitimate slow diagonal

Lemma 3.1 (Countable slow diagonal). *Suppose $E_j(x) \geq 0$ and $E_j(x) \rightarrow 0$ as $x \rightarrow \infty$ for each fixed $j \in \mathbb{N}$. Then there exists a nondecreasing integer function $J(x) \rightarrow \infty$ such that*

$$\max_{1 \leq j \leq J(x)} E_j(x) \rightarrow 0. \quad (3)$$

No quantitative lower bound for the growth of $J(x)$ follows.

Proof. For each m , choose X_m increasing so that

$$\max_{1 \leq j \leq m} E_j(x) \leq \frac{1}{m} \quad (x \geq X_m).$$

Set $J(x) = m$ for $X_m \leq x < X_{m+1}$. □

Enumerate any countable collection of fixed primitive pairs and fixed cutoffs. Apply [lemma 3.1](#) to the errors in (2), normalized by $\log \omega(x)$. This produces some slowly expanding frozen family on which the logarithmic conclusion is uniform. It does *not* prove that the prescribed TPC family lies inside that diagonal. In particular, it gives no allowable power law for as , R , the interval origin, or the number of outer keys.

4 Two exact Tauberian firewalls

4.1 One terminal logarithmic value is insufficient

For a finite sequence $b_1, \dots, b_N$, define

$$A(N) = \sum_{m \leq N} b_m, \quad L_r(N) = \sum_{m \leq N} \frac{b_m}{r+m}, \quad r \geq 0.$$

Proposition 4.1 (Exact translated Tauberian identity). *For every $N \geq 1$,*

$$\boxed{A(N) = (r+N)L_r(N) - \sum_{m=1}^{N-1} L_r(m).} \quad (4)$$

Proof. In the right side, the coefficient of b_k , for $k < N$, is

$$\frac{r+N}{r+k} - \frac{N-k}{r+k} = 1,$$

and the coefficient of b_N is also 1. □

Thus a terminal bound for $L_r(N)$ does not control $A(N)$ without the complete path $L_r(m)$ and its signed average. This is the exact anchored defect emphasized in TPC-96 [\[3\]](#).

4.2 A reciprocal window can cancel while its mass does not

Proposition 4.2 (Finite-window separation). *Let $\Omega \rightarrow \infty$, and on $(x/\Omega, x]$ define*

$$b_m = \begin{cases} -1, & x/\Omega < m \leq x/\sqrt{\Omega}, \\ +1, & x/\sqrt{\Omega} < m \leq x. \end{cases}$$

Then, up to integer endpoint errors,

$$\sum_{x/\Omega < m \leq x} \frac{b_m}{m} = O(\Omega/x), \quad (5)$$

$$\sum_{x/\Omega < m \leq x} b_m = x \left(1 - \frac{2}{\sqrt{\Omega}} + \frac{1}{\Omega} \right) + O(1). \quad (6)$$

In particular, reciprocal cancellation has no formal implication of ordinary cancellation on one terminal window.

Proof. The two reciprocal masses are

$$\log \frac{x/\sqrt{\Omega}}{x/\Omega} = \frac{1}{2} \log \Omega, \quad \log \frac{x}{x/\sqrt{\Omega}} = \frac{1}{2} \log \Omega,$$

with harmonic-sum endpoint error $O(\Omega/x)$. Counting the two interval lengths gives (6). $\square$

The sequence in [proposition 4.2](#) is an analytic counterexample, not a claim about Liouville signs. It proves that the desired Tauberian transfer cannot follow from the norm and window information alone.

5 Actual H3 quantifiers

After TPC-127 and one TPC-128 CRT component, a literal inner sum has the schematic but fully typed form

$$\sum_{t \in I_f} \lambda(q_f t + r_f - 2) \lambda(q_f t + r_f) \rho_f(t) W_f(t) e^{-2\pi i \beta_f t}, \quad (7)$$

where $q_f, r_f, I_f, \rho_f, W_f, \beta_f$ all belong to the predeclared actual family at scale X . The mask ρ_f retains any remaining literal periodic and coprimality factors.

Definition 5.1 (Callable actual-family prefix theorem). A theorem is callable by the termwise H3/zero-mode route only if, for a fixed $\delta > 0$, it proves

$$A_f^{(0)} := \sum_{t \in I_f} |\rho_f(t)|,$$

and, for every component with $A_f^{(0)} > 0$,

$$\max_{f \in \mathcal{F}_X} \max_{T \in I_f} \left| \sum_{\substack{t \in I_f \\ t \leq T}} \lambda(q_f t + r_f - 2) \lambda(q_f t + r_f) \rho_f(t) e^{-2\pi i \beta_f t} \right| \leq X^{-\delta + o(1)} A_f^{(0)}, \quad (8)$$

with one uniform $o(1)$, followed by a separately proved BV/outer-envelope estimate for W_f . Equivalently, a direct weighted theorem may retain W_f inside the sum, but then it must use the literal comparison mass

$$A_f^{(W)} := \sum_{t \in I_f} |\rho_f(t) W_f(t)|$$

and all required prefixes. Zero-mass components have identically zero sums and are discharged before taking either normalized maximum.

[Theorem 1.1](#) does not prove (8): its coefficients are fixed, its weight is reciprocal, its conclusion is $o(\log \omega)$ rather than a fixed power, and it has no stated uniformity over the growing actual family. The slow diagonal of [lemma 3.1](#) supplies no prescribed growth rate and therefore does not repair this mismatch.

6 Claim boundary and STOP rule

Statement	Level and status
Tao fixed-form terminal logarithmic theorem	Proved external theorem with native quantifiers.
Fixed finite-mask corollary (2)	Unconditional frozen-family result; auxiliary L1.
Countable slow diagonal	Exact non-effective lemma (L0).
Tauberian identity and window separation	Exact/elementary firewall (L0).
Actual growing-family prefix theorem (8)	Not proved (L2 only if established).
Physical H3 or positive zero-mode exponent	Not proved.

STOP. If an argument cites only (1) or (2) but needs any of growing modulus, prescribed cutoff growth, ordinary weight, actual phase, all prefixes, or uniform outer reassembly, it stops at the firewall. Calling the slow diagonal “the physical family” without proving containment is also forbidden.

GO. A future proof may proceed if it establishes (8) on the exact actual family, or a direct weighted substitute with an explicit exponent and return ledger. That would be new arithmetic information.

TPC-129 therefore records a real unconditional shadow of the route while preventing it from being misreported as fixed-2 physical progress. It proves no parity breach, prime-pair lower bound, Hardy–Littlewood asymptotic, or twin-prime theorem.

References

- [1] Terence Tao. The Logarithmically Averaged Chowla and Elliott Conjectures for Two-Point Correlations. *Forum of Mathematics, Pi*, 4:e8, 2016. doi: 10.1017/fmp.2016.6.
- [2] Liang Wang. Squarefree Masks in the Shift-Two Frame, 2026. Preprint.
- [3] Liang Wang. Anchored Logarithmic Primitives on Translated Fibers. https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-96-anchored-tauberian-translation-defect, 2026.